

[Quiz 1 (SOLUTION): January 27, Spring 2025]

## CSE 309: Theory of Automata

1. (10 points) Assume  $\Sigma = \{a, b\}$ . Give two strings that are members and two strings that are not members of following languages.
- (a)  $a(ab)^*$
  - (b)  $\Sigma^*a\Sigma^*b$
  - (c)  $(aa \cup ab \cup ba \cup bb)^*a$
  - (d)  $(\varepsilon \cup a \cup b)^*b$
  - (e)  $\{w : w \in \Sigma^*, |w| \bmod 2 = 0 \text{ and } |w| \bmod 3 = 0\}$

**Solution:**

|     | Members                              | Non-members            |
|-----|--------------------------------------|------------------------|
| (a) | a, aab, aabab                        | $\varepsilon$ , b, ba  |
| (b) | ab, aab, babb                        | $\varepsilon$ , a, b   |
| (c) | aaa, aba, baa, bba, aaabbba          | $\varepsilon$ , ab, bb |
| (d) | b, aaaab, bbbbbb                     | $\varepsilon$ , a, aba |
| (e) | $\varepsilon$ , ababab, aabbabababaa | a, ab, bab             |